

ESO208 COMPUTATIONAL METHODS FOR ENGINEERS

ASSIGNMENT 4

MAXIMUM MARKS: 30

DUE DATE: 5 SEPTEMBER 2026

Q.1 [20 marks] Consider a system of equations $AX = B$ in which

$$A = \begin{bmatrix} 7 & 2 & -3 \\ 2 & 5 & -3 \\ 1 & -1 & -6 \end{bmatrix}$$

(a) Write a code to perform LU decomposition of matrix A. [5]

(b) Use LU decomposition to solve the system of equations for

$$[B] = \begin{bmatrix} 12 \\ 18 \\ -6 \end{bmatrix} \quad [5]$$

(c) Use LU decomposition to calculate the inverse of A. [5]

(d) Calculate the condition number for matrix A using the row-sum norm. [5]

Q.2. [10 marks] Write a code for calculating the Cholesky decomposition of the following matrix

$$\begin{bmatrix} 8 & 20 & 15 \\ 20 & 80 & 50 \\ 15 & 50 & 60 \end{bmatrix}$$